

STATEMENT OF WORK

Project Title

Silver Branch Plant Survey 3

General Information

1.0 Background and Purpose

The purpose of the contract is to survey specified areas of the Ottawa National Forest for the presence of Regional Forester's Sensitive plants, State of Michigan listed plants, other plants of potential viability concern, priority non-native invasive plants, and plants of cultural significance (see lists in attachments). Surveys of the selected areas will be conducted one or two times. All 350 stands (7,992 acres) must be visited at least once. In addition, at least 105 stands (30%), totaling at least 2,398 acres (30%) must be visited a second time. All surveys should occur between **May 15 and September 15, in 2026 and/or 2027**. At least 3 weeks must pass between consecutive surveys in a stand. All surveys must be completed by September 15, 2027. All results and completed paperwork must be submitted by October 15, 2027. The Contractor shall progressively accomplish the work at a rate that will result in completion of the contract within the time specified. The Contractor shall furnish all labor, equipment, supervision, transportation, supplies, and incidentals (except those designated as Government-furnished property listed in Section G).

The surveys are for the proposed "Silver Branch Vegetation Management Project," named after local landmarks within the large project area, including the Silver River, and the North and South Branches of the Paint River. Survey results will be used to help develop project proposals and protection measures. This is the third plant survey contract for the Silver Branch project, following surveys completed in 2024 (Branch Botany Survey Contract) and 2023 (Silver River).

All surveys will occur on the Ottawa National Forest, in Houghton, Baraga, and Iron Counties. Stands selected for surveys are shown on the attached maps. Bidders are urged to inspect the bid items and work areas prior to submission of a bid. It is impossible to describe fully the area, work problems, and other conditions that may be factors in submitting a bid. Before any contract work is started, the Contracting Officer's Representative (COR) will arrange for a pre-work conference to discuss contract provisions and specifications.

Cover types included in the bid items are shown in Table 1.

Table 1. Survey stands cover types.

Cover type	Name	Acres	Percent
1	Jack pine	98.0	1.2%
2	Red pine	536.6	6.7%
3	Eastern white pine	120.8	1.5%
5	Hemlock	310.4	3.9%
11	Balsam fir-aspen/paper birch	444.0	5.6%
12	Black spruce	165.7	2.1%
16	White spruce & balsam fir	21.5	0.3%
18	Mixed swamp conifer	1077.3	13.5%
20	Northern hardwoods - hemlock	146.5	1.8%
48	Jack pine-oak	48.3	0.6%

71	Black ash-American elm/red maple	97.0	1.2%
76	Red maple (wet site)	25.1	0.3%
79	Mixed lowland hardwoods	15.8	0.2%
80	Sugar maple-northern red oak	34.3	0.4%
81	Sugar maple - yellow birch	492.7	6.2%
85	Sugar maple	301.7	3.8%
89	Mixed upland hardwoods	578.8	7.2%
91	Quaking aspen	1928.9	24.1%
92	Paper birch	213.3	2.7%
93	Bigtooth aspen	331.3	4.1%
95	Aspen-white spruce/balsam fir	1003.9	12.6%
	TOTAL	7991.9	100.00%

2.0 Definitions

COR – Contracting Officer’s Representative. The Contracting Officer will designate a Contracting Officer’s Representative at the time of award.

Contractor Requirements

3.0 Technical Requirements/Tasks

The Contractor and the Contractor’s employees (if applicable) must be able to identify all the species of vascular plants on the included list of rare plants and the included list of invasive plants of the Ottawa National Forest. (The Regional Forester’s Sensitive Species list for the Ottawa National Forest includes some bryophytes and lichens; these are not required survey species under this contract.) These specifications will pertain to all items contained within this contract.

The Contractor will complete the following tasks:

3.1 Pre-field review

3.1.1. The Contractor shall review topographic maps, aerial photographs, compartment maps, stand type and other information prior to field work to select the most likely rare plant habitat areas within the survey stands. Species abstracts (e.g., from Michigan Natural Features Inventory at <https://mnfi.anr.msu.edu/publications/abstracts>); floras; and other relevant information should be used to determine which listed species are likely to occur in the survey areas.

3.1.2. The Contractor shall select access routes and check with Contracting Officer’s Representative (COR) if there are access concerns. Access to the survey stands may often be difficult. There are few roads open to Off Highway Vehicles (OHVs) within the survey area. Highway M64, Old M64, Forest Road 360, South Boundary Road, and the North Country Trail are the main access routes within the survey area.

3.1.3. Cross-country OHV travel is not permitted on the Ottawa National Forest. The Motor Vehicle Use Map (MVUM) for the Ottawa shows roads that are open for use by vehicle class; contractors are expected to follow this map unless negotiated otherwise with the District Ranger through the COR for

administrative use of selected roads. Requesting this use (e.g. for ATV) may be desirable to shorten some hiking distances.

3.2 Field surveys

3.2.1. The Contractor shall determine habitat relationships and phenology for each of the taxa on the included lists (162 rare plants, 38 invasive plants, and 36 plants of cultural significance). Survey most intensively within the project area stands where and when the species under consideration are likely to be found. Searching for rare plants is the highest priority; record the invasive and cultural plants if encountered. The Contractor shall make a good faith effort to distribute searches throughout all potential community types or habitats where rare plant taxa may be found, and to do so at the times of the year appropriate to identify the plants. A good faith effort may require two field visits to the mesic areas in order to cover the various blooming periods (phenology) of vascular plants in the survey timeframes described above.

3.2.2. The Contractor will conduct surveys on a controlled intuitive or meander basis. That is, crisscross the stand while detouring to the most likely habitats for listed plants, such as rich forest areas, inclusions of rock outcrops, wetlands, etc. The Contractor will record survey routes with a handheld GPS unit (using datum NAD83) and submit routes electronically in a format compatible with Ottawa GIS and computer systems (format will be discussed at pre-work meeting). Ideally each GPS track record will correspond with one day.

3.2.3. All survey stands must be surveyed at least once between May 15 and September 15. In addition, the Contractor will survey at least 40% of awarded stands and 40% of awarded acres a second time. The contractor will choose what stands to survey a second time, and document their reasoning in their stand tracking notes. Criteria for surveying a stand twice include:

- Higher floral diversity.
- Presence of rare plants in the survey stand.
- Unusual habitats.
- Rock features.
- The first survey was early or later in the survey window, and may have missed the best time to find target rare species.
- Duff layer shows unusual lack of impacts from invasive earthworms.

3.2.4 If a rare plant population is located, information about the plant identity, number of individuals, precise location, area of coverage by population, and habitat characteristics will be recorded. For species that are RFSS, P-RFSS, Ottawa NF Extirpated, or Michigan-listed (99 species), complete a Michigan Natural Features Inventory Special Plant Survey Form. For species on the Rare List that are not RFSS or state-listed (63 species), less detail is needed. For those populations, record GPS location, date, observer, area, at least one photo, and notes on abundance and habitat; submit by email, Word document, spreadsheet, or links to observations on iNaturalist. **Field notes or field-completed paper forms must be transcribed by the contractor onto the MNFI fillable (pdf) form for submission to the Ottawa NF, and should be submitted electronically to the COR.** These forms will be shared with MNFI for state-listed species. Vouchers and/or photographs will be collected as appropriate and will be given to the Forest. Provide forms by 9/30 of the year the site was discovered.

3.2.5. Rare plant locations must be recorded using a GPS unit in **NAD83 decimal degrees** format (handheld, readily available GPS unit; survey grade is not required but can be used). Locations of the

previously documented rare plant populations in the project area will be provided to the contractor at the pre-work meeting and are not included in the solicitation. Known rare plants (and lichens) in the project area include *Artemisia ludoviciana*, *Botrychium michiganense*, *Calypso bulbosa*, *Carex backii*, *Clematis occidentalis*, *Dryopteris fragrans*, *Dryopteris goldiana*, *Hydrophyllum virginianum*, *Hypotrachyna revoluta*, *Muhlenbergia uniflora*, *Panax quinquefolius*, *Phegopteris hexagonoptera*, *Piptatheropsis canadensis*, *Pyrola americana*, *Thelypteris noveboracensis*, *Usnea longissima*, *Viola novae-angliae*, and *Zizania palustris*.

3.2.6. If a population of an Ottawa non-native invasive plant (NNIP) species of concern is found, information about the plant identity, number of individuals, location, infestation extent and habitat characteristics will be recorded. Contractor may use the Ottawa National Forest Weed Reporting form (see attachments). Alternatively, an Excel spreadsheet with all the required fields on the Weed Reporting form may be used. See included NNIP list for reporting requirements by invader species including which species require locations to be recorded using a GPS unit. Provide forms or spreadsheet with NNIP finds by 9/30 of the year the site was discovered. Locations of known priority NNIP are shown on the compartment maps with this solicitation. Known infestations include garlic mustard, wild chervil, glossy buckthorn, exotic honeysuckle, purple loosestrife, wild parsnip, Burnet-saxifrage, Scots pine, Japanese knotweed, giant knotweed, and crown vetch.

3.2.7 If a population of a plant of cultural concern is found, collect a GPS point according the directions for each specified species. Note that several of these species are common, and we only seek a few locations with abundant plants. The contractor may often stop after five locations or record the species as present in the strand tracker notes. See separate "Instructions for Reporting Plants of Cultural Significance."

3.2.08 During field surveys, record species encountered, notes for each stand, and notes on plant habitats, to be recorded in final report and stand tracking spreadsheet as specified below.

3.3. Provide at the **pre-work meeting** a survey plan, to include approximate dates for surveys and survey locations; approximate acreage to be covered daily; surveyor names; vehicle make, model and license number; and so on. Vehicle information is needed in case of emergency and because the public may report the same vehicles being found frequently in a given area to our law enforcement officers.

3.4. Report survey progress to the COR on a bi-weekly basis by e-mail or telephone. Have surveyed routes available for review during contract inspections, to demonstrate stand survey coverage.

3.5. Complete the Forest data sheets, stand tracking form, and a final report as directed in the protocol: final copies typed; electronic versions of the forms will be provided to the selected contractor and **electronic form submission is required**, including noting the presence of unusual communities. The weed reporting form is optional and may be completed by hand and then scanned into a pdf file for submission while the other forms should be typed.

3.6. For locations of plants on the 2026 Ottawa "rare plant list" (included in solicitation) that are RFSS, P-RFSS, Ottawa extirpated, or Michigan-listed, complete the Michigan Natural Features Inventory (MNFI) special plant forms to provide data on population size, vigor, threats, etc. Associated species should be listed by scientific name to species epithet, not to genus unless distinguishing features (flowers/fruits) are lacking. Non-vascular plants may be listed as "moss", "lichen" if no further identification can be made. Directions should be sufficient so that the site could be relocated, including description of landmarks that could be used in relocation (see below for monumenting directions). Habitat description

should be detailed, conveying a mental image of the features rather than “rich hardwoods” or similar brief descriptions. Completion of the form includes providing a copy of a topographic map with population location shown. Final copies of the forms must be typed into the MNFI fillable (pdf) form. Blank forms are available from the MNFI website at <https://mnfi.anr.msu.edu/species/report>, and a master copy will be provided to the contractor. Rare plant discoveries should be reported to the COR within 2 weeks, along with a GPS reading for the site and number of individuals. The MNFI form does not have to be completed within the 14 days, although it is often easiest to complete a draft field form when the population is found.

3.7. If the rare species population can support it (based on Contractor's professional judgment), the Contractor will collect and prepare one good quality voucher specimen (unmounted) **with label** for submission to the Forest (it will be forwarded to the University of Michigan Herbarium). Labels must be **typed** and follow standard format (a sample label is shown below). If the contractor holds a Michigan collection permit, include the number on the label and provide a **copy of the permit** with the season's vouchers. Refer to Voss 1999 (The Michigan Botanist 38:57-63) for more information on label preparation. Label width is no more than approximately 4 inches wide.

PLANTS OF THE OTTAWA NATIONAL FOREST (MICHIGAN) <i>Penthorum sedoides</i> L. Gogebic Co.: T48N R44W Section 23 Common on edge of backwater in Presque Isle River at Highway 28 crossing, southeast side. With <i>Scirpus cyperinus</i>, <i>Polygonum</i> sp., <i>Acer rubrum</i> seedlings. Fruits reddish. A.B. Smith 557 12 July 1999
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3.8. Mark rare plant locations in the field with color-coded flagging provided to the Contractor, in a visible but discreet manner.

3.9. For small populations, where, in the Contractor's professional judgment, it would be difficult to relocate the population using handheld GPS readings (accurate to about 50 feet under tree canopy) and description of landmarks, complete an additional monumenting process as follows. Locate three live and healthy-appearing witness trees, at least 5 inches in diameter at breast height, close to the rare plant population but in varying directions from the population. Record the species and dbh of these witness trees. Then measure and record the distance from the population to each of the three trees, and the bearing from the population to the tree. For example, the site record might read “8 inch red maple at 34 feet and 70 degrees, 12 inch white spruce at 25 feet and 180 degrees, 5 inch balsam at 15 feet and 290 degrees”. At the base of the three witness trees, nail a metal tag in the direction facing the population. Tags and nails will be supplied to the contractor. (Contractor must supply compass and measuring tape.)

3.10. Provide 3-5 good quality, higher resolution digital photographs of each rare plant taxon located, with 2-3 showing the plant and diagnostic characters in detail and 1-2 showing the plant in its habitat.

Provide photos in .jpg format **at resolution about 2400x3000 pixels**. Provide additional photos if the identity of the plant is enigmatic (e.g. *Botrychiums*). Smart phone cameras may not have the resolution needed; contractor will have to check their phone or tablet's settings and may need to carry a separate camera to take photos at 7-10 mp especially for difficult species where close focus is needed.

3.11. Notify the COR of any high priority or new invader weed locations within 2 weeks of observation (unless they are already documented, and information on documented sites will be provided to Contractor at pre-work meeting).

3.12. A stand tracking form shall be completed for each bid item, showing dates surveyed and habitat notes for **all** stands included in the contract. See attachments for a sample form. This must be submitted in **electronic form as an Excel spreadsheet** (the sample in the solicitation attachments is a 2-page pdf to show the data fields; the Excel version will be provided to the selected contractor at the pre-work meeting). **This is the most important data form (other than MNFI special plant forms); data expectations for this form will be discussed at the pre-work meeting.** Since it must be typed, rows and columns can be made larger by the contractor to accommodate lengthy comments. See separate "Stand Tracker Instructions."

3.13. Prepare one comprehensive species list for all vascular plants observed under this contract. This replaces the "Plant Survey Species Form" we previously required for each habitat type. Include the scientific and common name for each species. Score the abundance of each species as follows: A for abundant, C for common, F for frequent, O for occasional, T for trace. Sort the species list by trees, shrubs/subshrubs/vines, forbs, graminoids, and ferns & allies. The species list should include identification of grasses and sedges to species. Listing plants to the genus level is only acceptable when flowers or other necessary features were not found during the course of the contract. Additional notes for each species are optional but could include collection numbers or details on where uncommon plants were observed. Listing bryophyte, lichen, and fungal species is optional. Submit as either a Microsoft Word form or Excel spreadsheet.

3.14 Prepare one final report for the contract. The final report must include:

- A summary of the area surveyed.
- Discussions of each habitat encountered, following "A Field Guide to the Natural Communities of Michigan" by Cohen et al. (2015).
- A section on rare plants found
- A section on invasive plants.
- A section on plants of cultural concern.
- A section on conservation and management recommendations for the project area.
- A section on other observations, such as wildlife, historic sites, dump sites, or signs of improper human activity (see 3.15 below).

This final report replaces the "Plant Survey General Form" required with previous contracts. Final report must be submitted as a Microsoft Word document. See Protocol for Rare Plant Surveys for additional information.

3.15. During the performance of this contract, the Contractor is requested to contact the COR to report any unusual sightings such as large stick nests, artifacts or heritage sites (that are not already marked with flagging or in obvious ways), or other special features. Also include such observations in the final report.

4.0 Deliverables / Schedule

Key Deliverables

Item No.	Deliverable	Reference Tasks	Delivery Format	Due By
1	Survey plan	3.3	Verbal	At pre-work meeting.
2	Surveyor names and vehicle info	3.3	Verbal	At pre-work meeting.
3	Stand Tracker spreadsheet.	3.12	Microsoft Excel.	10/15/2027
4	Species list	3.13	Excel or PDF	10/15/2027
5	Final report	3.14	Word	10/15/2027
6	NNIP reports	3.2.6	Excel or PDF	10/15/2027
7	Special Plant (MNFI) forms.	3.2.4 & 3.6	PDF	10/15/2027
8	Plant vouchers with typed labels.	3.7	Printed labels, Word, or PDF	10/15/2027
9	Digital photos of special plants and habitats.	3.10	JPG	10/15/2027
10	GPS tracks for survey routes	3.2.2	Shapefile, text file, or GPX	10/15/2027
11	Reports of rare plants	3.6	Email or phone	Within 2 weeks of observation.
12	Reports of previously unmapped high priority or new invader weeds	3.11	Email or phone	Within 2 weeks of observation.
13	Reports of other items of note and general check-ins on progress	3.4 & 3.23	Email or phone	Biweekly when working on this contract

5.0 Government-Furnished Property

The Government will provide the following items of Government property to the Contractor for use in the performance of this contract. Items to be returned to the Government are stated below.

- 5.1 Ottawa National Forest map.
- 5.2 Motor Vehicle Use Map Ottawa National Forest East.
- 5.3 Compartment/stand maps for the bid items (same as attached in solicitation).
- 5.4 Topographic maps, 1 copy of each quad covering the bid items. Fourteen maps total: Beechwood, Covington, Echo Lake, Gibbs City, Glitter Lake, Lake Mitigwaki, Marten Lake, Perch Lake, Porter Lake, Sidnaw, Sunset Lake, Tunis, Vista Falls, and Winslow Lake.
- 5.5 Aerial photography and terrain mapping (lidar) for the survey stands if Contractor wants this.
- 5.6 Shapefiles or img files of stand outlines to upload to GPS units (we have tested this process with various Garmin units, but do not know if it will work with other brands of mapping GPS units). Alternatively, centroids for survey stands can be provided for uploading to GPS units.
- 5.7 Flagging to mark rare plant locations (return excess at end of contract).

- 5.8 Tags and nails for monumenting small populations (return excess at end of contract).
- 5.9 Master paper copies of the data sheets to be used, which may be photocopied by Contractor for use in the field, as well as the electronic data forms for final reporting.
- 5.10 List and map of documented rare plants and NNIP in project area.
- 5.11 A tablet pre-loaded with survey stands and other layers (such as roads and ownership) in the Field Maps application. Field Maps will allow the Contractor to enter Stand Tracking data in the tablet if they choose. Field Maps will also allow the Contractor to access three offline base maps: Ottawa National Forest Visitor Map, ESRI topo map, and a modest-resolution aerial photo. The best use of the tablet may be for navigation and aerial photos.

6.0 Travel

N/A

7.0 Contractor's Key Personnel

Contractor must provide sufficient personnel to complete all technical requirements/tasks and deliverables within the contract period. Refer to sections 3 and 4 for details.

8.0 Security Requirements

N/A

9.0 Data Rights

Data collected and reports prepared under this contract are the property of the USDA Forest Service. Data publication is prohibited without prior consent of the USDA Forest Service, Ottawa National Forest and the USDA Forest Service reserves the right to edit any potential publications. The USDA Forest Service will be acknowledged in any publications using such data.

See also: "Data Rights – General" in Section C

10.0 Section 508 - Electronic and Information Technology Standards

N/A for contractor. If deliverable, electronic documents under this performance work statement are not Section 508 compliant, it will be the Government's responsibility to make them so.

11.0 Performance Requirement Measures

11.1 Inspection

Inspection will be done as work progresses to ensure compliance with contract terms, conditions, and specifications. This will include the COR, or alternate COR, joining the Contractor periodically during field surveys and reviewing field data sheets and GPS routes of completed surveys. Contractor should be prepared to show coverage of stands and may submit completed GPS routes electronically to the COR prior to the inspection, in order for the COR to make a map showing coverage, if that is helpful to Contractor. Any work not being properly performed will be brought to the immediate attention of the Contractor. The Contractor shall take prompt corrective action for any deficiencies or deviations that may occur under this contract.

After the Contractor informs the Government that work has been completed, the Government will perform a prompt final inspection and advise the Contractor of acceptance or rejection of the work performed.

11.2 Acceptance

Acceptance of work for all items requires that all contract requirements are met including field reporting sheets are completely filled out, species identifications fall within known or likely distribution ranges (or vouchers are collected), and any voucher specimens are correctly and completely prepared and labeled. Submitted GPS routes should show passage through all stands. If original inspection results are unacceptable, the Contractor must continue work until results are acceptable or request a re-examination. This request must be made in writing within 5 days of notice of unacceptable work.